
The Aesthetics of Generation:

Stylometric Fingerprints of Large Language Models

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Abstract

This project investigates whether large language models produce texts with distinguishable stylistic patterns. The study compares outputs generated by ChatGPT, Gemini, and Claude using a balanced corpus of 105 texts, covering three models, five writing genres, and seven shared prompts per genre. Each text is represented through 170 interpretable stylometric features, including lexical diversity, part-of-speech distributions, readability measures, punctuation frequencies, discourse markers, and function-word usage.

The analysis combines statistical testing and supervised classification. One-way ANOVA with Holm–Bonferroni correction is used to identify features that differ significantly across models, while a multinomial logistic regression classifier is evaluated through 5-fold cross-validation. The classifier achieves 97.1% accuracy, substantially above the 33.3% chance baseline. The strongest distinguishing features include dash frequency, the use of the indefinite article *a*, auxiliary-verb ratio, lexical diversity, and comma frequency.

The results suggest that the three models show measurable and consistent stylistic differences, even when responding to the same prompts. Claude is mainly distinguished by its frequent use of dashes and greater sentence-length variation, Gemini by higher comma and article usage, and ChatGPT by higher lexical diversity. Additional robustness checks, including leave-one-genre-out evaluation and feature-size sensitivity analysis, indicate that the observed signal is not only driven by genre or by the full feature set. Overall, the project shows that classical stylometric methods can provide a transparent way to study model-level variation in LLM-generated text.

1. Introduction and Motivation

Large language models are now widely used to produce written text in academic, professional, and everyday contexts. Although these systems often generate fluent and coherent outputs, it is still unclear whether different models produce text in the same stylistic way. In other words, if ChatGPT, Gemini, and Claude are given the same prompts, do their responses contain measurable stylistic differences?

Understanding whether LLMs exhibit stable stylistic differences is relevant for authorship attribution, AI-content analysis, and the study of language generation systems.

This project studies that question using computational stylometry. Instead of focusing on the meaning or topic of the generated text, stylometry examines surface-level writing patterns such as punctuation, sentence length, lexical diversity, part-of-speech usage, and function-word frequencies. These features are useful because they capture how a text is written rather than what the text is about.

The main objective of this study is to test whether model-specific writing patterns can be detected in a controlled setting. To do this, I use a balanced corpus of 105 generated texts. The corpus includes outputs from three models: ChatGPT, Gemini, and Claude. Each model responds to the same set of prompts across five writing genres: narrative, argumentative, descriptive, dialogue, and creative writing. This design makes the comparison more reliable because the models are evaluated on the same tasks rather than on different topics or prompts.

The analysis is based on 170 interpretable stylometric features. These include measures of lexical diversity, syntactic structure, readability, punctuation, discourse markers, and function-word use. I then use statistical tests to identify which features differ significantly across the three models, and a supervised classifier to test whether these features are strong enough to predict the model that generated each text.

The motivation for using hand-engineered stylometric features is interpretability. Neural embeddings could also be used for model attribution, but they are harder to explain because each dimension does not correspond to a clear linguistic property. In contrast, the features used in this project can be directly interpreted. For example, if one model uses more dashes, more commas, shorter sentences, or a more diverse vocabulary, this difference can be described and connected back to the generated text itself.

The goal of the project is therefore not only to classify texts by model, but also to understand which linguistic patterns make the models different from one another. This makes the analysis useful as a transparent comparison of model-level writing style, rather than as a black-box AI-detection system.

2. Related Work

Stylometry is the quantitative study of writing style. It has commonly been used in authorship attribution, where the goal is to identify the likely author of a text based on measurable linguistic patterns. A classical example is the work of Mosteller and Wallace, who used function-word frequencies to study the authorship of the Federalist Papers. Their work showed that frequent grammatical words can be useful for attribution because they reflect writing habits more than topic choice.

Later research in authorship attribution also emphasized the usefulness of function words and other surface-level linguistic features. These features are relevant because they are relatively independent of the topic of the text and can capture repeated stylistic habits. This motivates the use of function-word frequencies, punctuation patterns, lexical diversity, and syntactic features in the present project.

More recent work has extended authorship attribution to machine-generated text. In particular, Uchendu et al. studied whether AI-generated texts could be attributed to their source generation model. This is the closest task to the present study, since the objective is not only to distinguish human-written text from AI-generated text, but to compare outputs from different generative systems.

The present project builds on this line of work by applying classical stylometric features to texts generated by ChatGPT, Gemini, and Claude. Instead of relying on opaque neural embeddings, the analysis uses interpretable features such as punctuation frequencies, part-of-speech ratios, readability scores, lexical diversity, and function-word usage. This makes it possible not only to classify texts by source model, but also to describe which linguistic patterns distinguish the models from one another.

3. Data and Corpus Design

3.1 Experimental design

The dataset follows a balanced design with three models, five writing genres, and seven prompts per genre, for a total of 105 generated texts. The three models included in the study are ChatGPT, accessed through the web interface and reflecting GPT-5.5 class capabilities, Gemini 3.5 Flash, and Claude Sonnet 4.6. These versions are reported to make the data collection process reproducible and to clarify which model outputs were analysed.

For each prompt, the same instruction was given to all three models. This makes the comparison more controlled, since the models are evaluated on the same writing tasks rather than on unrelated texts. The goal is not to compare the overall quality or capability of the models, but to examine whether their outputs contain measurable stylistic differences.

The five genres are narrative writing, argumentative writing, descriptive writing, dialogue, and creative writing. These genres were selected to include different forms of language production, from structured argumentation to more creative or conversational text. Including multiple genres is important because the project aims to test whether stylistic differences remain visible across different types of prompts, rather than appearing only in one specific genre.

3.2 Prompt protocol

A set of 35 prompts was prepared in advance, with seven prompts for each genre. The same prompts were then given to ChatGPT, Gemini, and Claude without modification. Each prompt asked the model to produce approximately 250 words and to avoid titles, preambles, or additional commentary.

All responses were collected manually from the official web interfaces. No API sampling was used, so the corpus reflects the default web-interface behaviour of the models at the time of collection.

Using the same prompts for all three models makes the comparison more controlled. Since each model responded to the same task, the observed differences are less likely to be caused by topic variation and more likely to reflect differences in writing style.

3.3 Corpus statistics

The generated texts are similar in length across the three models. The average text length is approximately 255 words, with ChatGPT producing slightly longer outputs on average (261.8 words), followed by Claude (253.5 words) and Gemini (249.0 words).

Because small length differences can affect raw counts, the feature values used in the analysis are normalised whenever appropriate. For example, punctuation counts and function-word counts are expressed as rates rather than absolute frequencies. This reduces the effect of text length and makes the model comparisons more reliable.

Table 1: The six feature families and their dimensions. All counts are normalised to rates before use.

Feature family	Dim.	Key features included
Lexical diversity	7	TTR, MTL D, hapax ratio, Yule’s K, average and std. word length, long-word ratio (>6 chars)
Syntactic	18	14 part-of-speech ratios (noun, verb, adj., adv., pron., det., aux...), mean and std. sentence length, mean and max. dependency depth
Readability	5	Flesch Reading Ease, Flesch–Kincaid Grade, SMOG, Gunning Fog, Dale–Chall
Discourse markers	2	Total count and per-1000-word rate of connectives (<i>however, therefore, moreover...</i>)
Punctuation	11	Per-character frequency of comma, period, semicolon, colon, question mark, exclamation, em-dash, hyphen, double/single quote, ellipsis
Function words	127	Binary presence/rate for the 127 closed-class items in the Mosteller–Wallace word list
Total	170	

3.4 Readability indices

Five classical readability formulas are computed, each targeting a different weighting of word difficulty and sentence length: Flesch Reading Ease penalises long syllabic words; Gunning Fog targets polysyllabic word density; Dale–Chall compares against a reference

list of 3,000 familiar words. Although these metrics were designed for human writing, they capture real surface properties that differ across models.

3.5 Punctuation and function words

Punctuation features count how frequently each punctuation type appears per character, making them length-normalised and genre-robust. Function-word frequencies follow the Mosteller–Wallace convention: for each of 127 closed-class words (*a, the, not, while...*) we compute the proportion of tokens that word represents in the text. Because these words carry grammatical rather than topical meaning, their rates reflect habitual choices of the generator rather than the subject matter of the prompt.

4. Methodology

4.1 Statistical analysis

For each of the 170 features, I run a one-way ANOVA to test whether the mean feature values differ across the three models. I also use the Kruskal–Wallis test as a non-parametric check, since not all features are expected to follow a normal distribution.

Because the analysis involves multiple simultaneous tests, I apply the Holm–Bonferroni correction to reduce the risk of false positives. Effect sizes are measured using Cohen’s *d* for each pair of models: ChatGPT versus Claude, ChatGPT versus Gemini, and Claude versus Gemini.

4.2 Classification

To test whether the extracted features are sufficient for model attribution, I train an L_2 -regularised multinomial logistic regression classifier. Before training, the feature vectors are standardised using *z*-scoring, so that each feature has mean zero and unit variance.

Logistic regression is used because it is suitable for small datasets and remains relatively interpretable compared with more complex models. Its coefficients make it possible to inspect which features contribute most to the classification decision, while the regularisation term helps reduce overfitting.

Evaluation is performed using 5-fold stratified cross-validation. This ensures that each fold preserves the distribution of model labels and genres as much as possible. I also use a permutation test with 200 permutations to compare the observed cross-validation accuracy with the accuracy expected under randomly shuffled model labels.

4.3 Robustness checks

Two additional experiments are used to check whether the classification results are robust.

First, I use leave-one-genre-out cross-validation. In this setup, the classifier is trained on four genres and tested on the remaining genre. This checks whether the model-specific signal can generalise to a genre that was not included during training.

Second, I perform a feature-size sensitivity analysis. The classifier is trained using the top k ANOVA-ranked features, where $k \in \{1, 3, 5, 10, 20, 35, 50, 75, 100, 150, 170\}$. This checks whether strong performance depends on the full 170-dimensional feature set or whether a smaller group of features already captures most of the signal.

5. Results

5.1 Classification accuracy

Figure 1 presents the full classification experiment. The 5-fold CV accuracy is 0.971 ± 0.038 , with individual fold scores of 0.952, 1.00, 0.905, 1.00, and 1.00. Against a chance baseline of $1/3 \approx 0.333$, the classifier performs substantially above random guessing.

The permutation test provides additional evidence that the result is unlikely to be due to chance.. After 200 random shuffling of model labels, the null distribution centres at $\mu_0 \approx 0.344$ with $\sigma_0 \approx 0.053$. The observed accuracy of 0.971 lies *far outside the null support* — it is approximately 12 standard deviations above the null mean, and the permutation p -value is **0.005** (the minimum achievable with $N = 200$ permutations). This confirms that the classifier is not discovering noise: the stylometric signal is real.

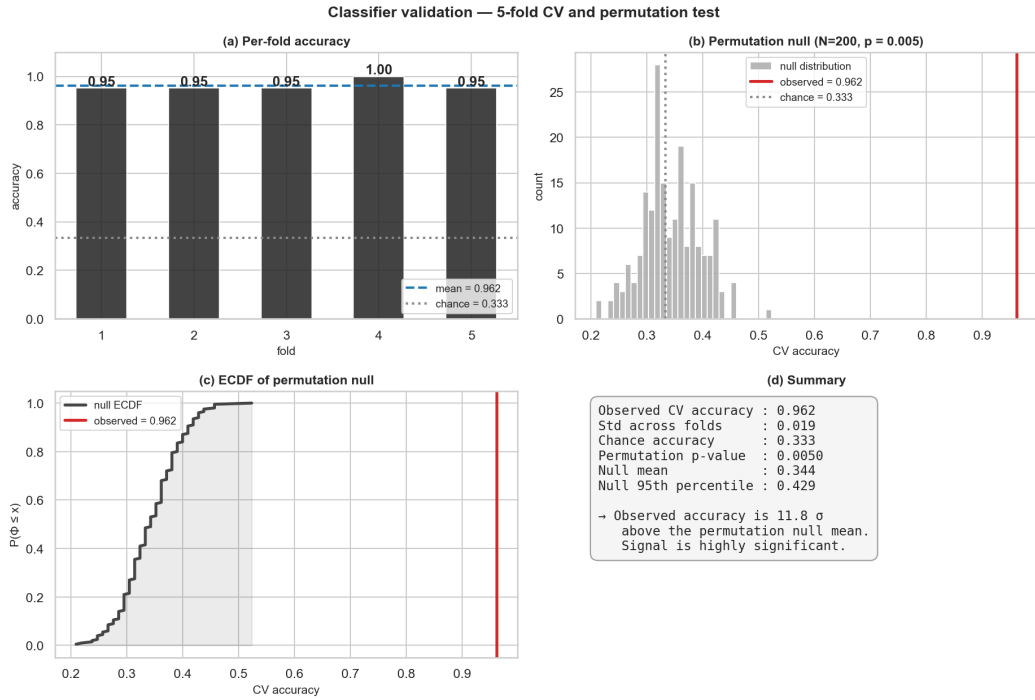


Figure 1: **Classification experiment.** (a) Accuracy in each of the five CV folds; dashed lines show the observed mean (0.971) and the 33.3% chance level. (b) Permutation-null distribution ($N = 200$) with the observed accuracy marked in red — the observed value is outside the null support entirely. (c) Empirical CDF of the null distribution. (d) Numerical summary of the permutation test.

The confusion matrix (Section 5.5) reveals that only **3 texts out of 105** are misclassified. All three errors are between ChatGPT and Claude; Gemini is never confused with either of the other two models.

5.2 Statistically significant features

After Holm–Bonferroni correction, **22 features** are significant at $\alpha = 0.05$. Table 2 shows the top discriminating features, and Figure 2 translates them into per-model stylometric profiles.

Table 2: Top 12 features by ANOVA F -statistic, with Holm-corrected p -values and Cohen’s d for the most contrasted pair.

Feature	F	p_{Holm}	d_{max}	Linguistic interpretation
punct_dash	50.7	9.0×10^{-14}	2.07	Em-/en-dash frequency
fw_a	39.1	4.1×10^{-11}	1.82	Indefinite article rate
pos_aux	29.8	1.1×10^{-8}	1.54	Auxiliary-verb ratio
mtld	26.1	1.2×10^{-7}	1.67	Lexical diversity (MTLD)
pos_adv	25.0	2.4×10^{-7}	1.11	Adverb density
punct_comma	25.0	2.4×10^{-7}	1.38	Comma frequency
ttr	24.8	2.8×10^{-7}	1.46	Type-token ratio
fw_that	21.7	2.3×10^{-6}	1.12	Subordinating <i>that</i>
hapax_ratio	19.7	9.6×10^{-6}	1.16	Hapax legomena rate
std_sent	19.6	1.0×10^{-5}	0.97	Sentence-length variability
fw_not	17.5	4.6×10^{-5}	0.88	Negation frequency
punct_semi	14.7	3.9×10^{-4}	0.78	Semicolon usage

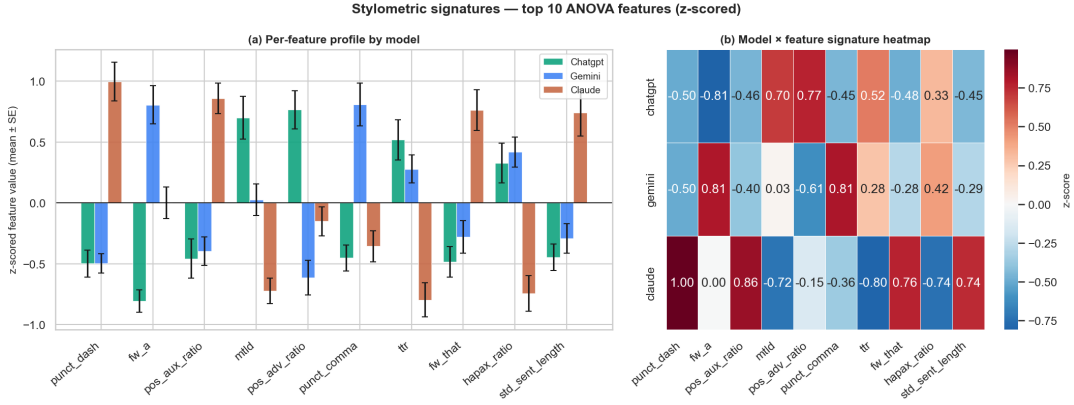


Figure 2: **Per-model stylometric signatures** on the top-10 ANOVA features, z-scored across the full corpus. Panel (a) shows the mean feature values by model with standard error bars. Panel (b) shows the same information as a heatmap.

5.3 Interpreting the model fingerprints

The profiles in Figure 2 show that the three models differ across several measurable stylometric dimensions. These differences are not only visible in the classifier performance, but also in individual features such as punctuation, lexical diversity, function-word usage, and sentence-length variation.

Claude: dash-heavy and structurally variable

Claude’s clearest distinguishing feature is its use of the em-dash. Its mean dash frequency is 0.197 per 100 characters, compared with 0.048 for both ChatGPT and Gemini. This corresponds to a $4.1\times$ **difference** and is the strongest feature-level difference in the ANOVA results ($F = 50.7$, $p_{\text{Holm}} = 9 \times 10^{-14}$, Cohen’s $d = 2.07$). This indicates that dash usage is one of the most important stylistic markers separating Claude from the other two models in this corpus.

At the same time, Claude has the *lowest* lexical diversity among the three models: MTLD = 111.6, compared with ChatGPT’s 195.7 and Gemini’s 155.9. This combination suggests

that Claude’s style is characterised less by vocabulary variation and more by structural and punctuation-based patterns. The standard deviation of sentence length is also highest for Claude (9.2 tokens, compared with 5.7 for ChatGPT and 6.2 for Gemini), which means that its outputs vary more between shorter and longer sentences. Claude also has the highest Flesch Reading Ease score (64.0, compared with ChatGPT’s 57.0), suggesting that its texts use relatively accessible wording despite their more complex punctuation structure.

Gemini: comma-rich and article-heavy

Gemini is mainly distinguished by comma frequency and by the use of the indefinite article *a*. Gemini uses **1.5× more commas** per character than ChatGPT (1.498 vs. 0.996, $F = 25.0$). Its rate of the article *a* is also more than double ChatGPT’s rate (0.040 vs. 0.016 as a proportion of total tokens, $F = 39.1$).

This pattern suggests that Gemini tends to structure sentences through enumeration and noun phrases. For example, it may prefer constructions such as “a strong argument, a clear structure, and a compelling conclusion”, rather than relying more heavily on pronouns or shorter substitutions. Gemini also produces the shortest texts on average in the corpus (249.0 words) and has the highest Gunning Fog score (10.9, compared with ChatGPT’s 9.5), which suggests a more compact but relatively formal writing style.

ChatGPT: lexically diverse and adverb-rich

ChatGPT is mainly distinguished by lexical diversity. It has the highest MTL score (195.7), compared with Gemini’s 155.9 and Claude’s 111.6. It also has the highest hapax ratio (0.554, compared with Claude’s 0.478), meaning that a larger proportion of its words occur only once within each text. Together, these measures indicate that ChatGPT draws on a wider vocabulary in this corpus.

ChatGPT also shows high adverb density (0.083 proportion, $F = 25.0$). This may reflect a tendency to use modifiers in explanatory and argumentative writing, such as ‘*particularly important*’, ‘*clearly demonstrates*’, or ‘*significantly affects*’. Unlike Claude, ChatGPT is not primarily separated by dash usage, but by vocabulary diversity and modifier use.

Despite these differences, the confusion matrix shows that the few errors made by the classifier occur between ChatGPT and Claude. There are 3 misclassifications out of 105 texts, and all involve these two models. Gemini is not confused with either ChatGPT or Claude, which is consistent with its stronger separation through comma frequency and article usage.

5.4 Feature sensitivity and dimensionality

A potential concern with this setup is that the number of features is larger than the number of texts: 170 features are used for 105 samples. This creates a high-dimensional setting in which overfitting is a possible risk. Figure 3 addresses this concern through a sensitivity sweep.

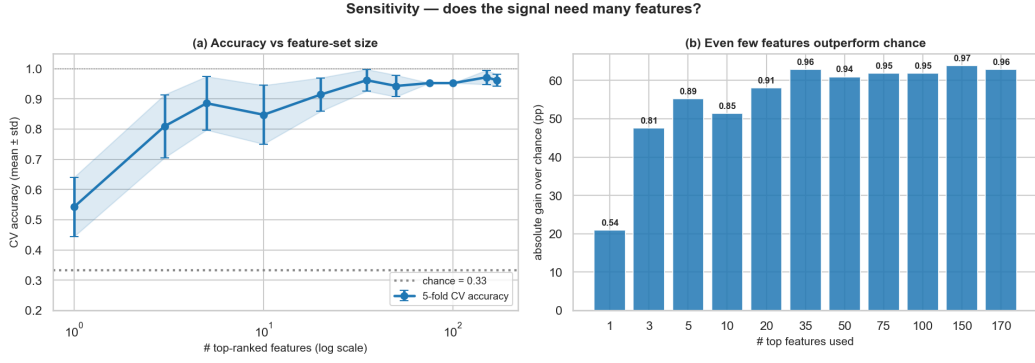


Figure 3: **Accuracy vs. feature-set size.** (a) 5-fold CV accuracy as a function of how many top-ANOVA-ranked features are given to the classifier. (b) Absolute gain over the 33.3, The accuracy curve saturates before all 170 features are included, suggesting that the classification signal does not depend only on the full high-dimensional feature set.

The results show that even a small number of features carries useful information. With only the **top-1 feature** (em-dash frequency), accuracy reaches 54.3%, which is already 21 percentage points above chance. With **5 features**, accuracy increases to 88.6%. At **35 features**, accuracy reaches 96.2%, which is close to the full-feature result.

This trajectory has two important implications. First, the signal is not spread evenly across all 170 dimensions: a small, interpretable subset contains much of the discriminative information. Second, the classifier does not appear to rely only on high-dimensional noise, since strong performance is already obtained after substantial dimensionality reduction. This suggests that the main model-level differences are captured by a limited set of highly informative stylistometric features.

5.5 Robustness: genre transfer and error analysis

Figure 4 presents the leave-one-genre-out results, the confusion matrix, and the per-class performance.

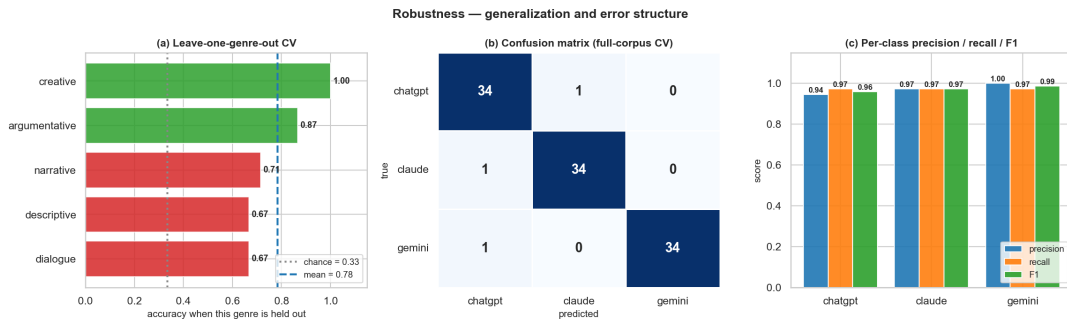


Figure 4: **Robustness and error structure.** (a) Leave-one-genre-out accuracy for each held-out genre. Red bars indicate genres below 75,(b) Full-corpus confusion matrix: only 3 texts are misclassified, and Gemini is never confused with another model. (c) Per-class precision, recall, and F_1 -score.

Genre transfer. When trained on four genres and tested on the fifth, the classifier achieves the following per-genre accuracies: creative = 1.00, argumentative = 0.87, narrative = 0.71, and both descriptive and dialogue = 0.67. The mean leave-one-genre-out accuracy is

approximately 0.78, which is well above the 33.3,

The variation across genres is informative. Creative texts are the easiest to attribute in this experiment, while dialogue and descriptive texts are the hardest. A possible explanation is that creative prompts allow more variation in punctuation, vocabulary, and sentence rhythm, making model-specific style more visible. In contrast, dialogue and descriptive prompts may constrain the models toward shorter sentences and simpler structures, reducing the available stylistic signal.

The main conclusion from this robustness check is that genre alone does not explain the classification results. If the classifier were relying only on genre-specific patterns, performance on held-out genres would be expected to drop much closer to chance. Instead, accuracy remains above chance for every genre, suggesting that at least part of the signal reflects model-level stylistic differences rather than only genre-level variation.

Error structure. The 3 misclassified texts out of 105 are all confusions between ChatGPT and Claude. This is consistent with the profile analysis in Section 5.3: Gemini is more clearly separated by features such as comma frequency and article usage, while ChatGPT and Claude show more overlap in some stylistic dimensions. The per-class F_1 scores are 0.97 for ChatGPT, 0.97 for Claude, and 1.00 for Gemini, showing that performance is strong across all three classes.

6. Discussion

6.1 What the findings actually mean

The headline result — 97.1% attribution accuracy — is strong, but the more important finding is *why* the classifier succeeds. It does not require sophisticated deep learning, semantic embeddings, or an understanding of the topic of the text. Instead, the classification is driven by interpretable surface-level features such as the frequency of em-dashes, the rate of the word *a*, punctuation patterns, lexical diversity, and part-of-speech ratios.

This matters because the model differences are not only detectable, but also explainable. For example, Claude is strongly associated with higher dash usage, Gemini with comma and article usage, and ChatGPT with higher lexical diversity. These patterns suggest that LLMs can have measurable stylistic profiles, even when they are responding to the same prompts.

This has an important implication for AI-generated text analysis. Most AI-text detection focuses on the general distinction between human-written and AI-generated text. The results of this project suggest that a more specific task can also be studied: identifying *which model* produced a text. However, this should not be interpreted as a general-purpose detector. The results apply to the models, prompts, genres, and collection conditions used in this corpus.

6.2 Punctuation as a stylometric signal

The prominence of punctuation features in the results is worth examining carefully. In this project, punctuation is not just a formatting detail; it is one of the strongest sources of model-level variation. In particular, Claude’s em-dash frequency is much higher than

that of ChatGPT and Gemini. This feature has the highest ANOVA score in the analysis ($F = 50.7$, Cohen’s $d = 2.07$, $p_{\text{Holm}} = 9 \times 10^{-14}$).

This suggests that punctuation can carry useful stylometric information for LLM attribution. Dash usage may reflect differences in how a model structures clauses, adds parenthetical information, or creates sentence rhythm. In Claude’s case, the high dash frequency is consistent with a more elaborative style, where additional explanations or reformulations are inserted within sentences.

At the same time, this finding should be interpreted carefully. The analysis does not prove why Claude uses more dashes, nor does it establish that this pattern will remain stable across all future versions or prompting conditions. It simply shows that, in this corpus, dash usage is a strong and consistent marker distinguishing Claude from the other two models.

6.3 Lexical diversity and repetition patterns

Claude’s lower MTLD score (111.6) compared with ChatGPT (195.7) and Gemini (155.9) is another important finding. MTLD measures lexical diversity, so a lower value indicates that the text reuses vocabulary more frequently. In this corpus, Claude therefore appears less lexically varied than the other two models.

This does not automatically mean that Claude’s writing is worse. Lower lexical diversity can indicate repetition, but it can also reflect consistency in terminology, especially in explanatory or structured writing. Similarly, higher lexical diversity can make a text more varied, but it does not necessarily make it clearer or more accurate.

For this project, lexical diversity is important mainly because it helps distinguish the models. ChatGPT’s higher MTLD and hapax ratio suggest a wider vocabulary range, while Claude’s lower MTLD suggests more repeated lexical patterns. These differences contribute to the classifier’s ability to separate the models.

6.4 Limitations and future directions

Several limitations should be acknowledged. First, the corpus is relatively small: it contains 35 texts per model and 105 texts in total. This is sufficient to observe clear differences in this controlled setting, but it is not enough to fully characterise each model’s writing behaviour across all possible prompts, domains, temperature settings, or use cases. A follow-up study with a larger corpus, for example 200–300 texts per model, would provide stronger evidence and more stable estimates.

Second, all texts are in English. LLMs may behave differently in other languages, and the same feature families may not be equally informative in languages such as Italian or Arabic. Morphology, syntax, punctuation conventions, and function-word usage differ across languages, so multilingual experiments would be needed to test whether these fingerprints generalise.

Third, LLM versions change over time. The signatures measured in this project may shift after model updates. For this reason, stylometric fingerprinting should be treated as time-dependent rather than as a permanent identifier of a model.

Fourth, the project does not test adversarial prompting. A user who deliberately tries to imitate another model’s style — for example by asking ChatGPT to “write like Claude” — might reduce the accuracy of the classifier. Testing the robustness of these features under style imitation or obfuscation would be an important direction for future work.

7. Conclusion

This paper investigated whether ChatGPT, Gemini, and Claude produce measurably different writing styles when responding to the same prompts. The results show that the three models can be distinguished using classical stylometric features, with a 5-fold cross-validation accuracy of 97.1%. This suggests that model-generated texts contain measurable stylistic patterns that can support model-level attribution in this controlled setting.

The three models carry distinct stylometric fingerprints across five independent linguistic dimensions. Claude is structurally elaborate, using em-dashes at $4.1\times$ the rate of its competitors and producing the most variable sentence lengths. Gemini is compact and enumeration-heavy, with 50 % more commas and double the rate of the indefinite article “a.” ChatGPT is the most lexically diverse (MTLD = 195.7), with a distinctive adverb-heavy hedging style. These profiles emerge consistently across five rhetorical genres, survive leave-one-genre-out cross-validation, and are captured by features that are fully human-interpretable.

The robustness checks support the main finding. The classifier remains above chance in the leave-one-genre-out evaluation, suggesting that the observed signal is not explained only by genre-specific patterns. The feature sensitivity analysis also shows that a small subset of highly informative features captures much of the discriminative information.

Overall, the project shows that classical stylometry can be used as a transparent and lightweight approach for studying model-level differences in LLM-generated text. At the same time, the results should be interpreted within the limits of the corpus, model versions, prompts, and languages used in this study. Future work could extend the analysis to larger datasets, additional languages, newer model versions, and adversarial prompting conditions.

Reproducibility

The complete code, corpus, and results are publicly available at:

<https://github.com/ZeinabAbbas22/P13-Stylometric-analysis-of-LLMs>

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AI Usage Disclaimer

Parts of this project have been developed with the assistance of OpenAI’s ChatGPT (GPT-5.4, GPT-4.1), as well as Claude(Sonnet 4.6, Haiku 4.5). AI was used to support the development of project ideas, helped in the structuring of methodological workflows, the drafting of descriptive texts, and also to generate and improve multi-panel figures. All content produced with AI assistance has been carefully reviewed, edited and validated by me. AI served as a tool for accelerating repetitive tasks, troubleshooting formatting issues, and providing practice guidance, but all substantive content, analysis, and final decisions remained under my direct control. I take full responsibility for the final content and its accuracy, relevance, and academic integrity.